

Solutions for Practice Problems for Week 5

Operating and Systems Programming 2022/23

1. There are no restrictions on commands executed in kernel mode whereas operations which might affect security cannot be executed in user mode.
2. The following operations should be privileged:
 - Set value of timer.
 - Clear memory of other processes.
 - Issue a trap instruction. It must be possible to execute some trap instructions in user mode so that switches from user mode to kernel mode can happen.
 - Turn off interrupts.
 - Modify entries in device-status table.
 - Switch from user to kernel mode.
 - Access I/O device.
3. The kernel runs all the time, hence any leaked memory will never be recovered. If there is insufficient memory left, the OS will crash.
4. Some interrupts are typically disabled while interrupts are processed. A long interrupt processing time increases the danger of losing other interrupts.
5. The security and safety mechanisms enforced by the OS may be bypassed, resulting in illegal data access or system crashes.